



Quantum Face Recognition with Multi-Gate Quantum Convolutional Neural Network

B. Tech. Project Mid-Semester Evaluation

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Introduction & Problem Statement



Why it matters

Face recognition is used in banking, security and government services



Why a new approach

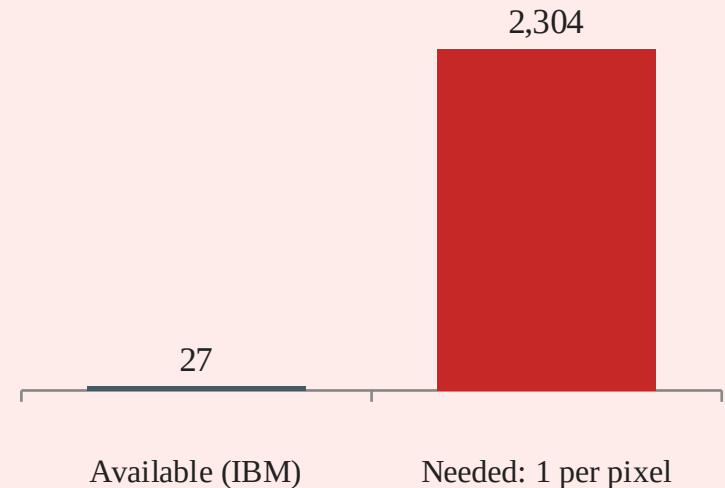
Deep CNNs are near-human (DeepFace: 97.35% on LFW) but need ever more compute



Why quantum

Quantum machine learning promises more efficient learning with far fewer trainable resources

The catch: qubits



Problem statement

Today's noisy quantum devices offer only 5–27 reliable qubits, but a 48×48 face image has 2,304 pixels. How can faces be recognised with a quantum model that needs only a handful of qubits, and how well does it work?

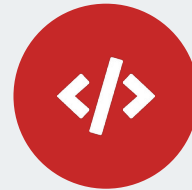
Objectives



1

Study quantum ML

Angle encoding, variational circuits, quanvolution



2

Implement the MG-QCNN

PennyLane + PyTorch, with an analytical model of the filter



3

Evaluate on ORL and Yale

10 random 70:30 splits, 1000 measurement shots



4

Analyse the design

Gates, entanglement, filter size, shot noise, hardware cost

Literature Review

Classical face recognition

- Gabor + KNN/SVM gave way to deep CNNs
- High accuracy, heavy computation
- Compute demand outgrows hardware

Variational circuits

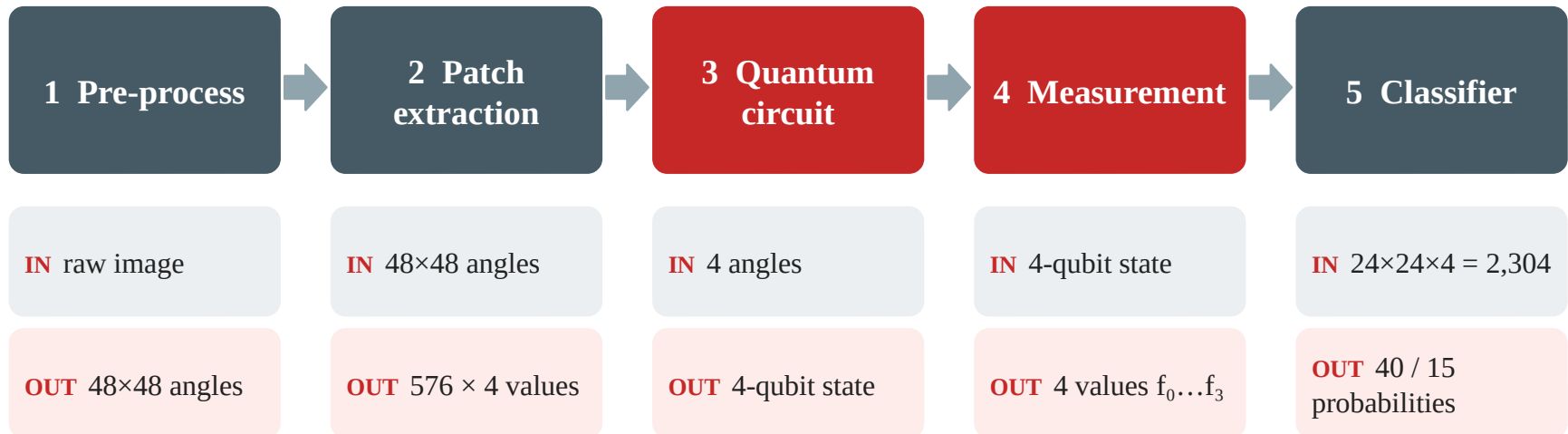
- Rotation angles act as weights
- Trained by gradient descent
- Barren plateaus favour small, fixed circuits

Quantum convolution

- Henderson: random, fixed filter
- Mattern: trainable but random
- Oh: fixed, one measured qubit
- MG-QCNN: fixed, trainable, all 4 qubits measured

Research gap: small datasets, noise-free simulation and no hardware test in the reference work. We add an independent implementation, an analytical model of the filter and a cost and noise analysis.

Proposed Methodology: Overview Architecture



*quantum part: 4 qubits, 8 trainable parameters
shared by all patches*

Tools

PennyLane (quantum circuit),
PyTorch (training); Kaggle
notebooks

Datasets

ORL: 400 images, 40 subjects ·
Yale: 165 images, 15 subjects

Evaluation

10 random 70:30 splits, 1000
measurement shots

Next: one slide per component, each with its input and output

Component 1: Data Pre-processing

INPUT

Raw gray image

ORL 92×112 · Yale 320×243

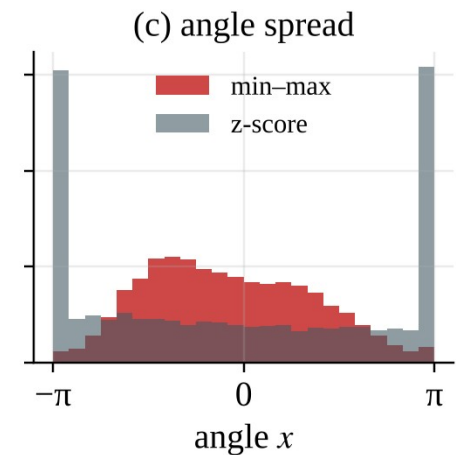
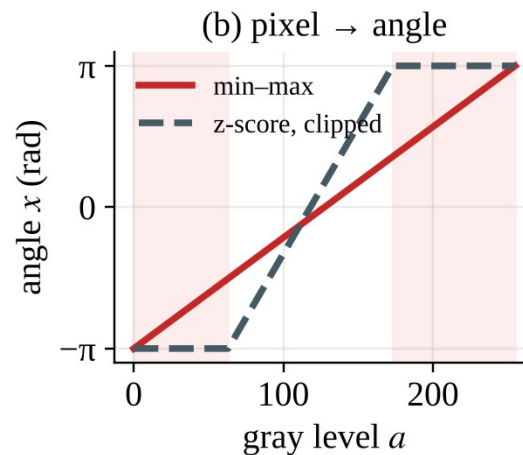
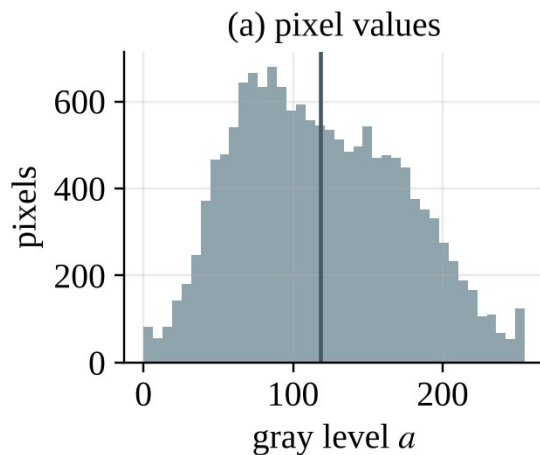
Pre-process

grayscale · resize · scale

OUTPUT

48×48 angle matrix

each pixel in $[-\pi, \pi]$



94.67%

ORL accuracy with min-max scaling

91.50%

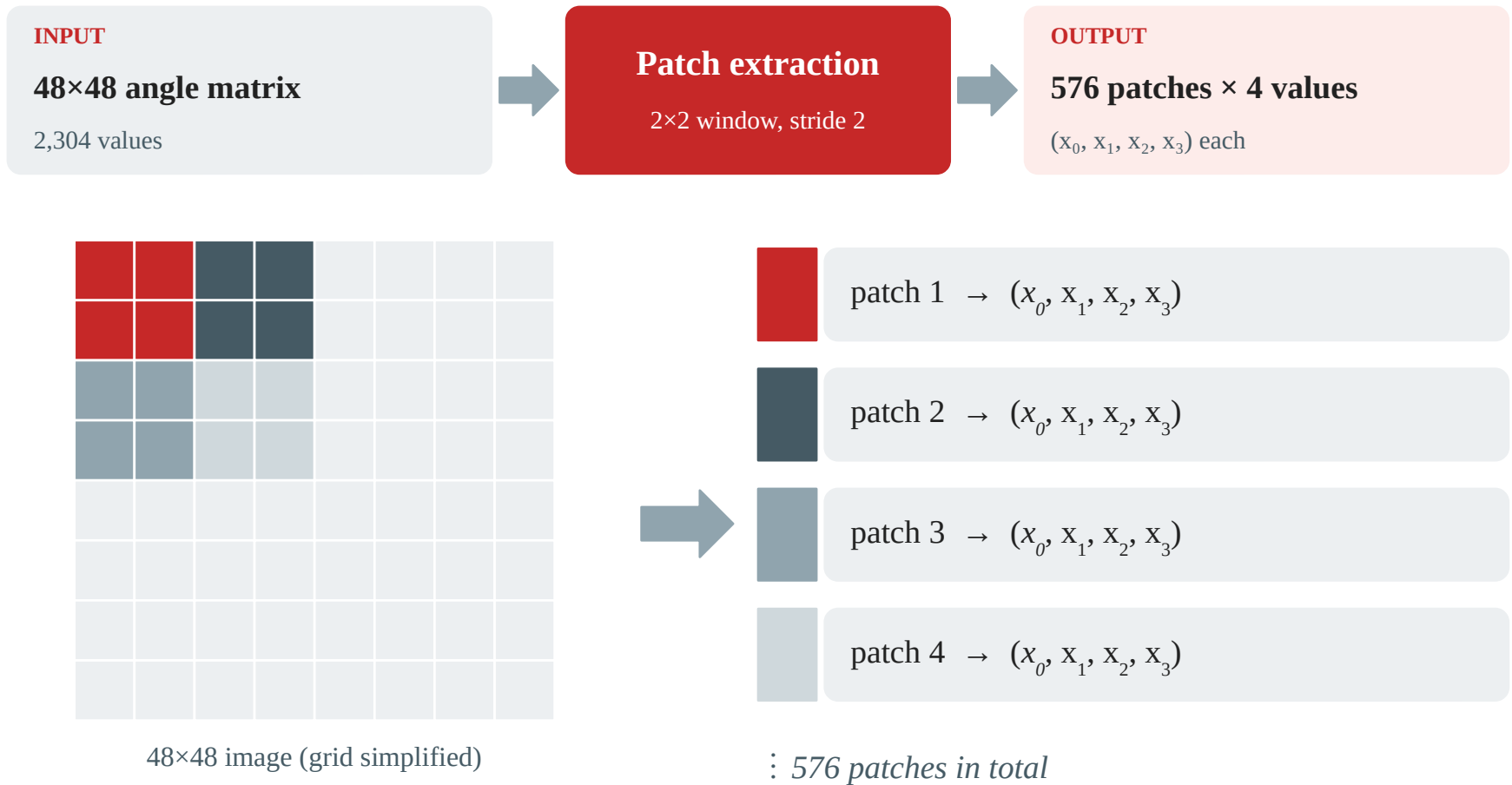
ORL accuracy with Gaussian z-score

10 / 10

seeds won by min-max (+3.17 pts)

Illustrative (synthetic gray levels): clipped z-scores pile about a third of pixels at $\pm\pi$.

Component 2: Patch Extraction



$24 \times 24 = 576$ patches. Each patch has 4 pixels, so 4 qubits are enough whatever the image size.

Component 3: Quantum Circuit

INPUT

4 angles $x_0 \dots x_3$

one patch

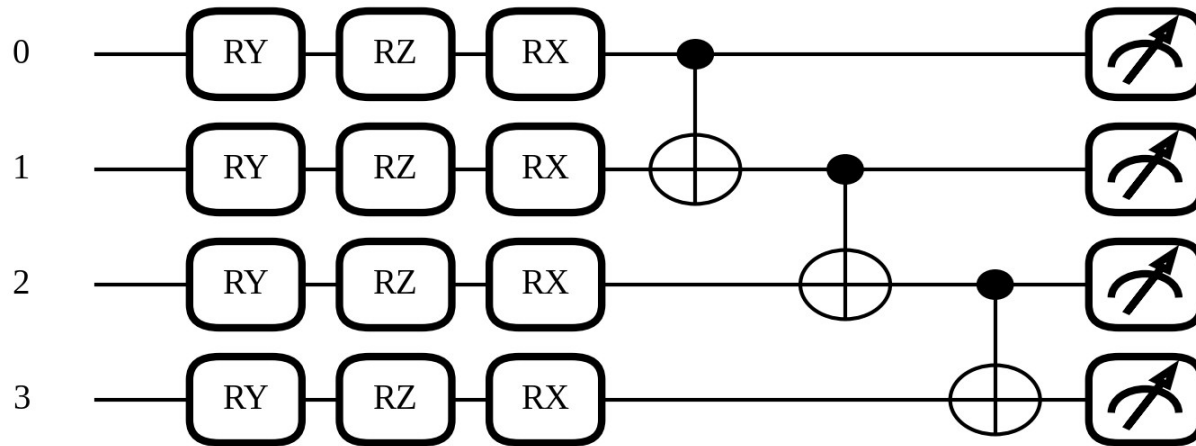
Quantum circuit

4 qubits · 15 gates

OUTPUT

Entangled 4-qubit state

depends on all 4 pixels



Encode

$R_Y(x_i)$ turns pixel i into a rotation

Train

RZ, RX: 8 parameters θ

Entangle

CNOT chain $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$

Component 4: Measurement

INPUT

Entangled 4-qubit state

from the circuit



Measurement

Pauli-Z on all 4 qubits



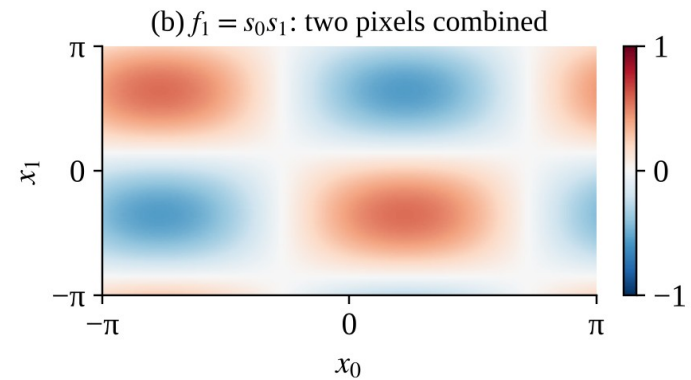
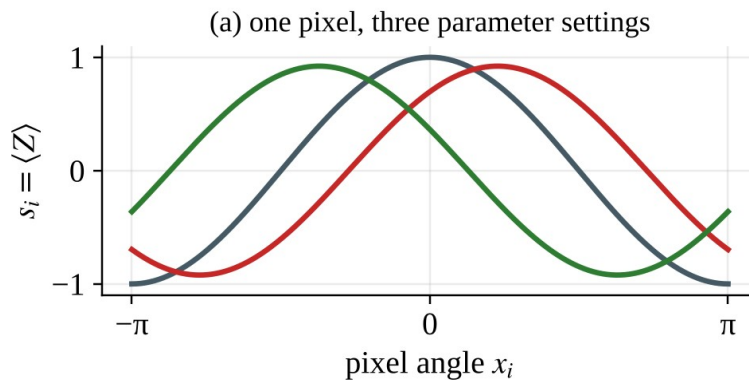
OUTPUT

4 values $f_0 \dots f_3$

each in $[-1, 1]$

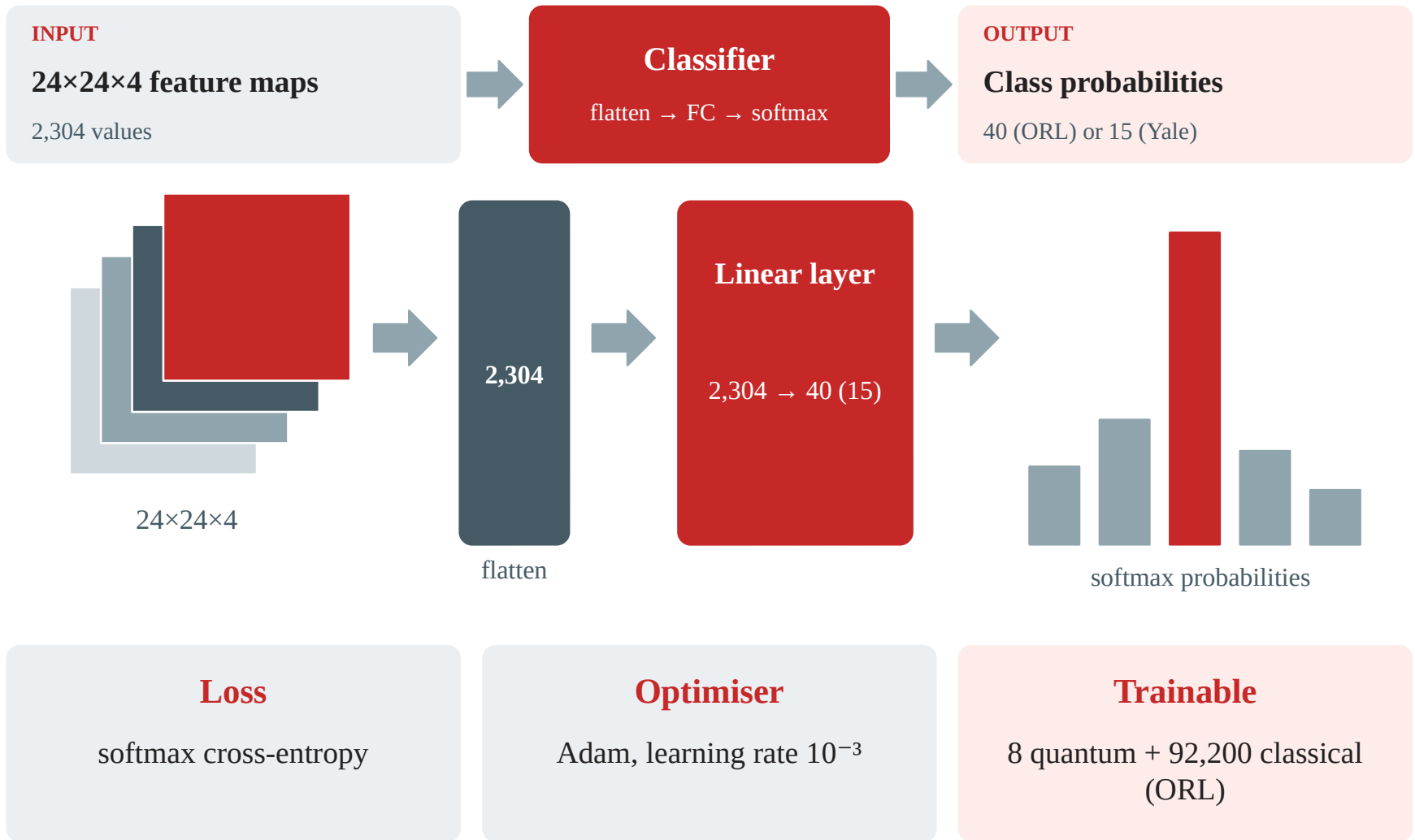
one qubit: $s_i = \cos \theta_{2i+1} \cos x_i + \sin \theta_{2i+1} \sin \theta_{2i} \sin x_i$

with CNOTs: $f_k = s_0 s_1 \dots s_k$ f_0 sees 1 pixel, f_3 sees all 4

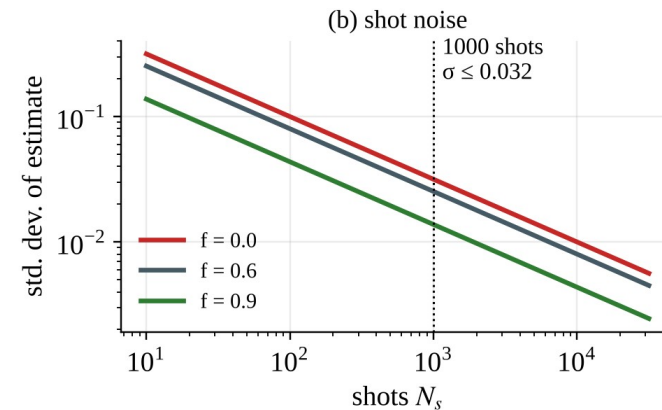
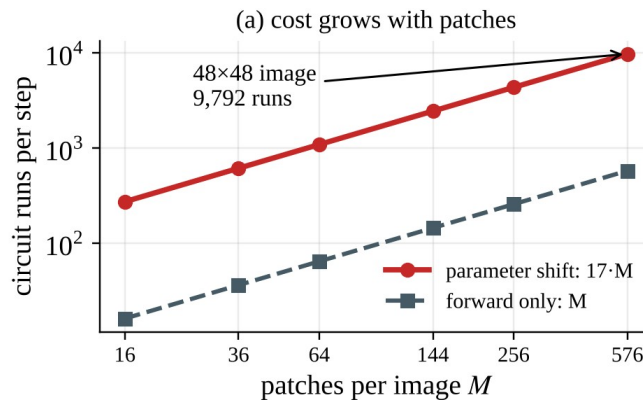


Verified: closed form matches the full simulation to 7×10^{-16} ; 1000 shots add at most 0.032 noise.

Component 5: Classifier



Training the Quantum Parameters



$$\partial f / \partial \theta = \frac{1}{2} [f(\theta + \pi/2) - f(\theta - \pi/2)]$$

parameter shift on hardware: two extra runs per parameter; backprop gives the same gradient in one pass

17

runs per patch

9,792

runs per image per step

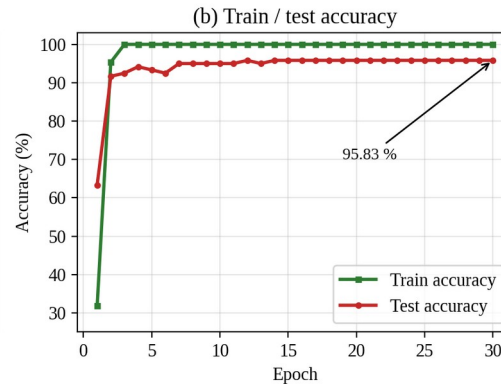
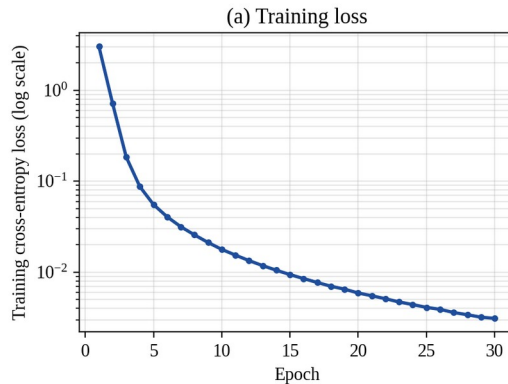
8.2×10^7

runs, one ORL training

2.2×10^{-15}

gradient mismatch

Results & Discussion: ORL

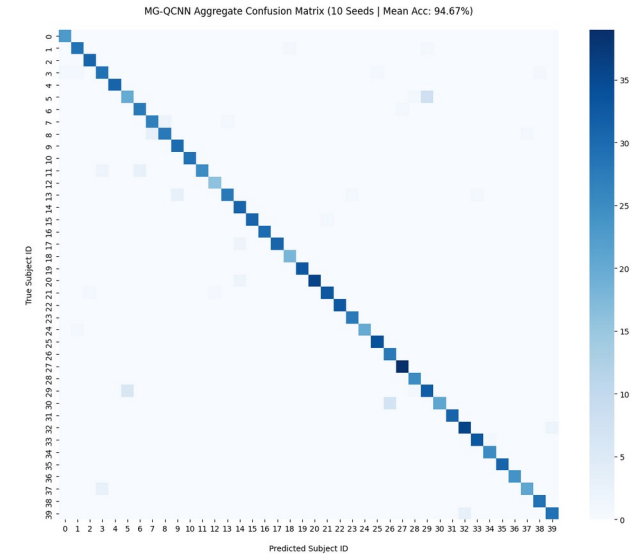


94.67%

mean test accuracy (max 97.50 · std 1.68)

100% → 95.83%

train from epoch 3 · test from epoch 14 (seed 0)

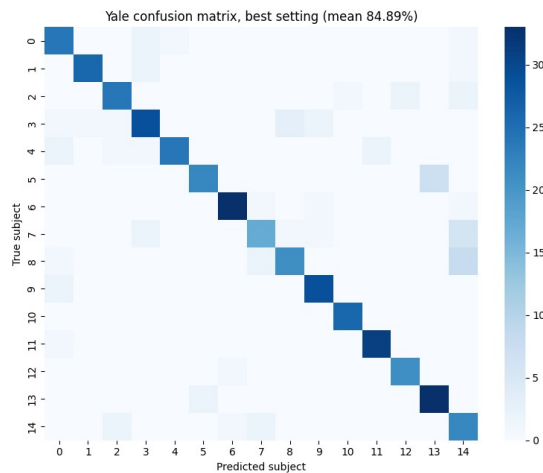
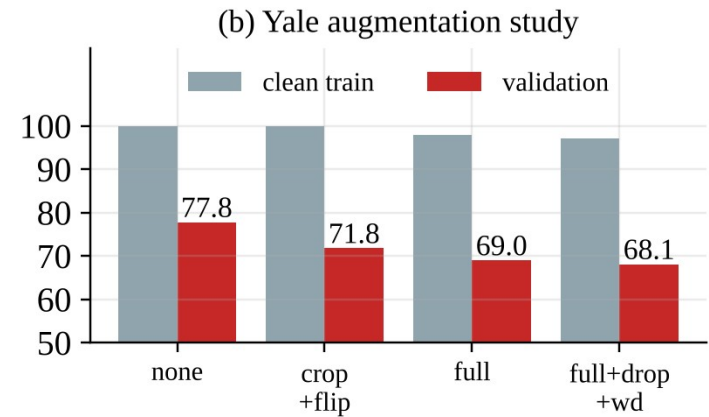
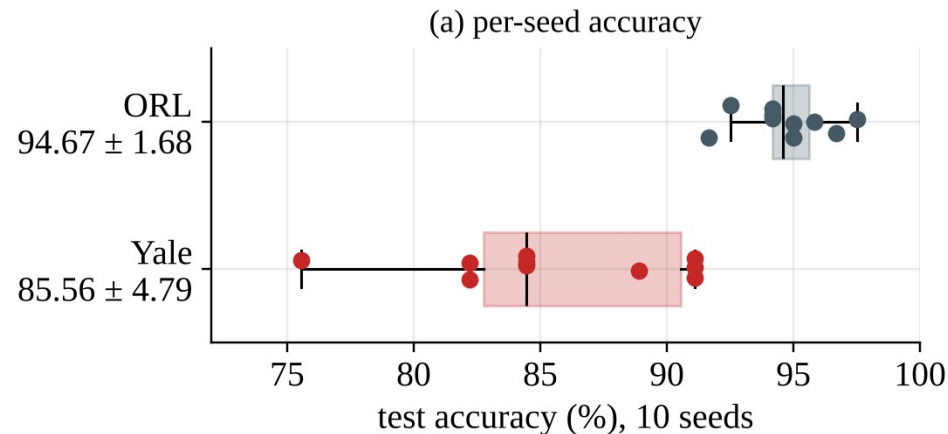


ORL confusion matrix, 10 seeds (1,200 predictions)

Discussion

- Learns fast: test accuracy 63% → 92% in epoch 2, above 95% from epoch 7
- Min-max scaling beats Gaussian on all 10 seeds (94.67% vs 91.50%)
- Errors are scattered single confusions; the 4.17-point train-test gap shows some over-fitting

Results & Discussion: Yale



85.56%

mean accuracy · max 91.11 · std 4.79

22 → 29 pts

train-validation gap with augmentation

- Over-fits: 100% train accuracy on only 120 images
- Augmentation did not help (validation 77.8 → 68.1%)
- One test image = 2.22 points, so Yale varies more than ORL
- Confusions: subjects 7, 8 read as 14; subject 5 as 13

Yale confusion matrix (tuning run, 450 predictions)

Future Work & Conclusion



Achieved

- MG-QCNN built in PennyLane + PyTorch
- Closed-form filter verified
- ORL 94.67%, Yale 85.56%
- Min-max beats Gaussian on all ORL seeds



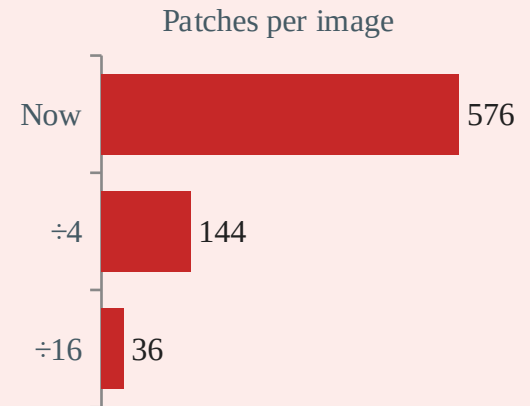
Limitations

- Noise-free simulation, not hardware
- Yale over-fits: 100% train vs about 85% test
- No ablation or noise study yet



Future work

Add a feature-extraction stage before the quantum layer to cut the patches needing circuit runs



8 quantum parameters · ORL 94.67% · Yale 85.56%

Fewer patches mean 4× to 16× fewer circuit runs and make real parameter-shift training practical.

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